TASK 2 · RESEARCH REPORT

LLM Prompt Engineering for RASE Analysis

§47 Aufenthaltsräume · Musterbauordnung (MBO)

1. Introduction

The objective of this task was to investigate whether a Large Language Model (LLM) can reproduce a human-generated RASE annotation of a legal regulation.

In Task 1, a manual RASE analysis of §47 Aufenthaltsräume from the Musterbauordnung (MBO) was created. The clause contains a combination of general requirements, special provisions for attic rooms, measurement logic, and explicit exceptions. The manual annotation served as the reference model for evaluating the performance of an LLM.

> **Main Research Question:** Can an LLM reproduce a human-level RASE annotation through iterative prompt engineering?

2. Manual RASE Analysis (Task 1)

Before evaluating the LLM, a manual RASE analysis was produced. The clause was interpreted as containing the following elements:

Category	ID	Description
Requirement	R1	Habitable rooms must have a clear ceiling height of at least 2.40 m.
Requirement	R2	Habitable rooms in roof spaces must have a clear ceiling height of at least 2.20 m over at least half of their net floor area.
Application	A1	All habitable rooms (Aufenthaltsräume).
Application	A2	Habitable rooms located in roof spaces (Dachraum).
Exception	E1	The requirements do not apply to residential buildings of Building Classes 1 and 2.

Important Human Interpretation

> *"Raumteile mit einer lichten Raumhöhe bis zu 1,50 m bleiben außer Betracht."*
>
> This sentence was **not** considered a standalone RASE element. Instead, it was interpreted as **measurement methodology** that explains how the floor area must be calculated. This interpretation became the most important evaluation criterion during Task 2.

3. Methodology

Three prompt versions were developed and tested against the manual RASE annotation. Each prompt used the same regulation text and became progressively more sophisticated.

Version	Purpose
V1	Basic RASE extraction — keyword-driven identification
V2	Legal interpretation — semantic and structural reasoning
V3	Human expert reasoning — lex specialis and hierarchy analysis

Outputs were evaluated on: correct RASE classification, legal interpretation quality, handling of exceptions, recognition of special rules, and similarity to the manual annotation.

4. Prompt V1 – Basic RASE Extraction

The first prompt instructed the LLM to identify Requirements, Applications, Selections, and Exceptions using the RASE methodology without any legal interpretation guidance.

Results

- ✓ **Requirement R1**
- ✓ **Requirement R2**
- ✓ **Applicability A1**
- ✓ **Applicability A2**
- ✓ **Exception E1**

Issues Found

Problem 1 – 1.50 m Statement

> The model classified *"Raumteile mit einer lichten Raumhöhe bis zu 1,50 m bleiben außer Betracht"* as a **Selection**, differing from the human interpretation where the sentence only describes the measurement procedure.

Problem 2 – No Lex Specialis Recognition

The model treated Sentence 1 and Sentence 2 as two independent requirements and did not recognise that the attic-room provision is a special rule replacing the general rule.

> **Evaluation:** Prompt V1 demonstrated strong extraction capabilities but weak legal reasoning. The model focused primarily on keywords rather than legal meaning.

5. Prompt V2 – Legal Interpretation

The second prompt introduced legal reasoning instructions, asking the model to think like a legal expert, explain ambiguities, analyse legal meaning, and distinguish between legal requirements and measurement rules.

Improvements

The output improved significantly. The model recognised that attic rooms represent a special case and identified the relationship between the two requirements, introducing the concept of lex specialis.

Remaining Problems

The 1.50 m statement was still treated as an Exception. Although the interpretation was better than in V1, the model still attempted to force the sentence into a RASE category.

Evaluation: Prompt V2 showed a clear improvement in legal reasoning and semantic understanding. However, the distinction between a true legal rule and supporting measurement logic remained unresolved.

6. Prompt V3 – Human Expert Interpretation

The third prompt incorporated expert review principles including legal hierarchy analysis, lex specialis reasoning, human-review guidance, and an explicit instruction that not every sentence must become a RASE element.

Results

Prompt V3 produced the closest match to the manual annotation:

- ✓ **The general rule correctly identified**
- ✓ **The attic-room special rule correctly identified**
- ✓ **The GK1/GK2 exception correctly identified**
- ✓ **The relationship between Sentence 1 and Sentence 2 recognised**
- ✓ **The legal hierarchy correctly mapped**
- ✓ **The 1.50 m statement identified as measurement methodology — not a RASE element**

Evaluation: Prompt V3 achieved **near-human performance**. The generated analysis closely matched the manual annotation produced in Task 1.

7. Comparison of Results

Feature	Human	V1	V2	V3
Requirement R1	✓	✓	✓	✓
Requirement R2	✓	✓	✓	✓
Applicability A1	✓	✓	✓	✓
Applicability A2	✓	✓	✓	✓
GK1/GK2 Exception	✓	✓	✓	✓
Lex Specialis Recognition	✓	✗	✓	✓
1.50 m Measurement Logic	✓	✗	✗	✓
Human-Level Reasoning	High	Low	Medium	High

8. Key Findings

Legal Interpretation is Harder than Extraction

The LLM could identify Requirements and Exceptions from the beginning. However, understanding the legal relationship between rules required multiple prompt refinements.

Measurement Methodology Causes Confusion

The most difficult sentence was: *"Raumteile mit einer lichten Raumhöhe bis zu 1,50 m bleiben außer Betracht."* The model repeatedly attempted to classify it as a RASE element. Human analysis showed it functions as measurement methodology rather than a legal requirement.

Lex Specialis Requires Explicit Guidance

Without additional instructions, the model could not recognise that the attic-room provision replaces the general requirement. Explicit legal reasoning guidance was necessary.

Prompt Engineering Has a Significant Impact

Each iteration improved the quality of the output. The transition from V1 to V3 demonstrated that prompt design is a major factor in achieving human-level results.

9. Conclusion

The experiment demonstrated that Large Language Models can perform RASE analysis with a high degree of accuracy.

Prompt	Key Outcome
V1	Successfully extracted the basic structure of the regulation but lacked deeper legal understanding.
V2	Improved semantic reasoning and identified the special legal relationship between the rules.
V3	Achieved near-human performance by incorporating expert interpretation principles and legal hierarchy analysis.

> The results suggest that LLMs can effectively support regulatory rule formalisation. However, **expert review remains necessary** to validate complex legal interpretations and resolve ambiguities.

10. Next Step – Task 3

Based on the results of Task 2, the next phase of the project is the development of an LLM-supported RASE extraction application.

The final prompt developed in Task 2 will be integrated into the application and tested against additional regulations and examples created by other groups.

> **Goal:** Evaluate whether the model can consistently reproduce human-generated RASE annotations across multiple clauses and whether prompt engineering can serve as a reliable foundation for automated rule formalisation.